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Appendix slides

What the main book could not hold: all 18 cases, the pattern matrix, UML designs, Project Hive WBS and RACI, the risk register, the board KPI scorecard, pending decisions, and media and certificate galleries.



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All 18 cases

#	Case	Domain	Headline results
01	Manufacturing AI built on an integrated core	Career highlights	+15% Yield; +18% OEE; −25% IT operating cost
02	Platform engineering with zero downtime	Career highlights	Zero downtime Core services; −30% Multi-cloud TCO; ×2 Release speed
03	AI governance and agents in production	Career highlights	+55% Automation efficiency; −40% Lead Time; −35% Defect escape
04	Listed biotech manufacturer | IT strategy and AI architecture	Enterprise AI platform	32 views; 4 quick wins; 100% GenAI rules aligned
05	Enterprise service-agent AI platform proposal	Enterprise AI platform	3 layers
06	Enterprise AI platform plan (RFP)	Enterprise AI platform	3 phases
07	Multinational apparel group | 2027 IT operating architecture	Manufacturing blueprints	6 gates; 4 architectures; 34 slide deck
08	Precision medical-device maker | cross-plant API platform and lakehouse	Manufacturing blueprints	6 stages; 26 week WBS; 4 sites
09	Multi-cloud and smart-manufacturing practice	Manufacturing blueprints	6 clouds; −30% cloud cost; +55% automation
10	Hospital-at-home AI decision portal	Industry AI cases	−20~30% 30-day readmission; −40% documentation time; 5 clinical scenarios
11	AI financial analysis and fraud simulation	Industry AI cases	5 fraud scenarios; STOP decision brake
12	Biopharma company | CISO 180-day resilience plan	Security & resilience	180 day plan; 14 day DFIR; 11 formal reports
13	Group CISO portfolio	Security & resilience	99.99% availability; −30% incidents
14	Current group | operations governance standard (incl. AI)	Governance & organization	17 L2 procedures; 11 L1 policies; 90 day plan
15	Zero-to-one organizational turnaround	Governance & organization	+0.1% DAU bet; 35 people
16	Game platform | internal-control handbook, nine cycles	Governance & organization	9 cycles
17	Frontier technology briefings	Research	5 issues; 50 page paper
18	Personal agent workspace and backup console	Agent tooling	13 directories; 7 day snapshots

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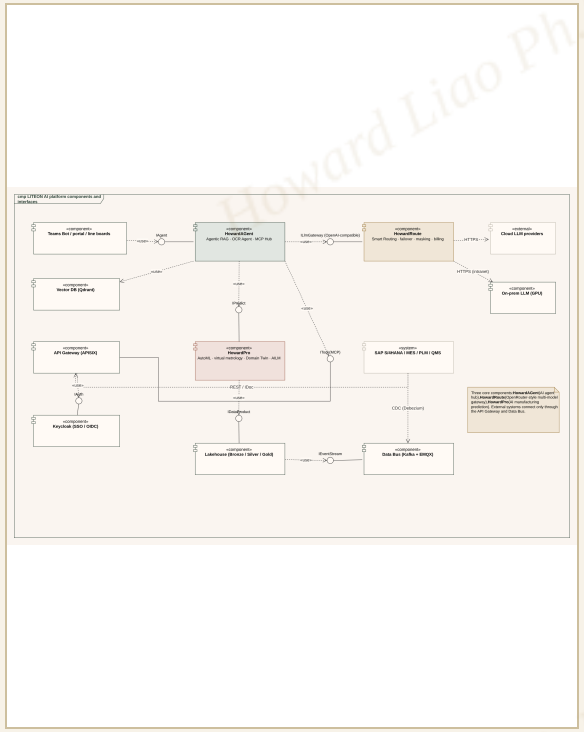
Patterns × cases: one backbone proven across industries

Pattern	Biotech AI	Service agents	AI platform RFP	Apparel group	Med-device plants	Multi-cloud mfg	Home care	Pharma security	CISO	Governance
Multi-model gateway and agents Gateway & orchestration	●	●	●			○	○			
Data bus and lakehouse Data bus & lakehouse	●			●	●	○	○			
Stage-gate delivery Decision gates				●	●			○		○
Human approval Human-in-the-loop	○	○	○				●			
AI governance and compliance ISO 42001 & regulation	●					●	●	●	○	●
Security and resilience Zero trust & HA	○					○		●	●	
FinOps and ROI Cost & value				●	○	●	○			

● Core design ○ Partly covered

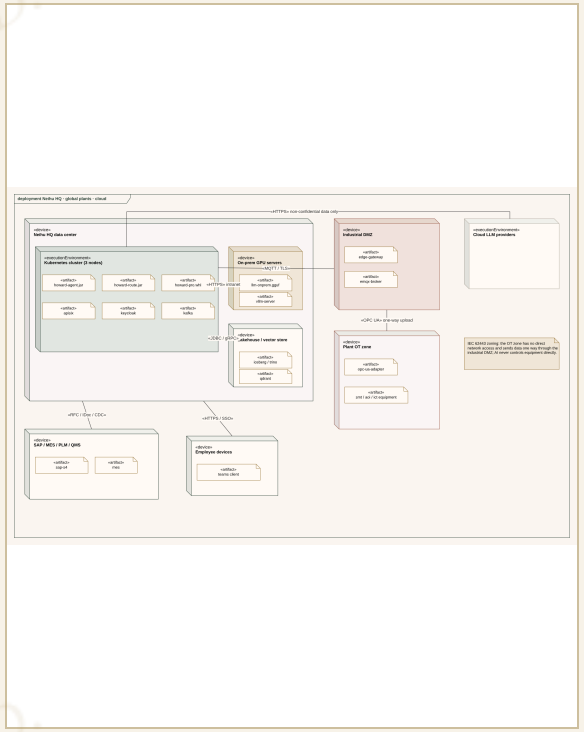
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Selected UML 2.x designs: from architecture to interaction



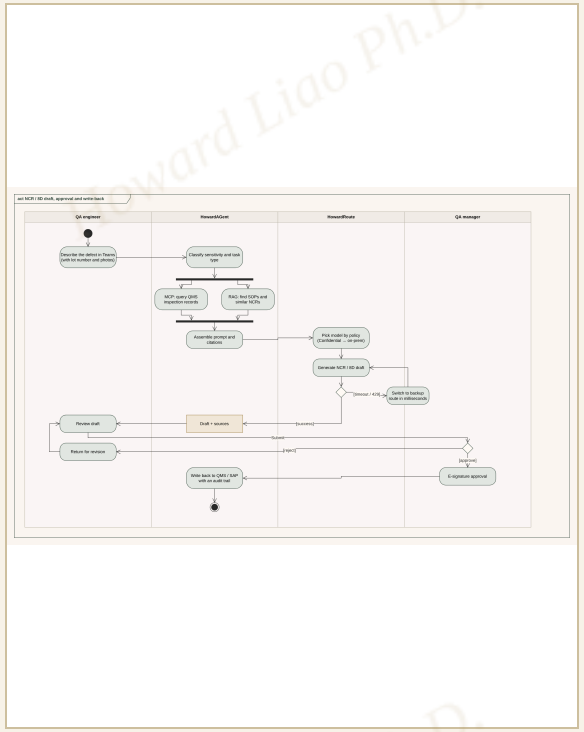
04 Component

Provided and required interfaces of the three cores and the data platform



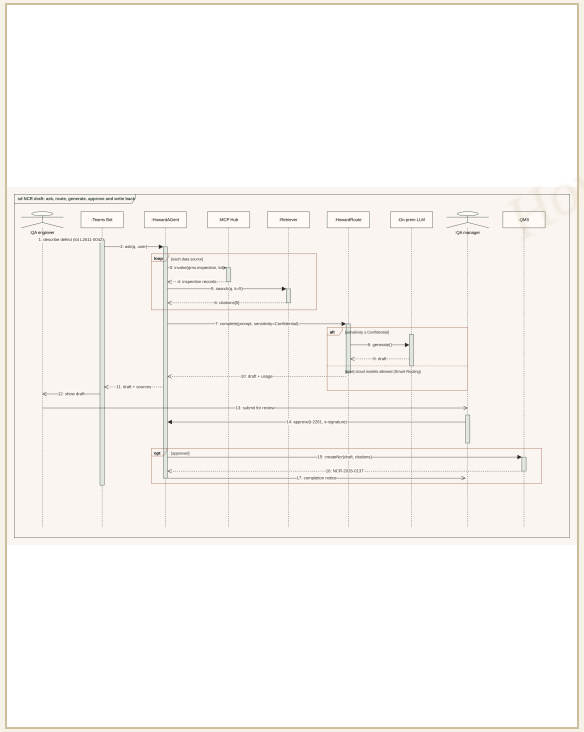
07 Deployment

Neihu HQ data center, on-prem GPUs, industrial DMZ, plant OT zone, cloud LLMs



09 Activity

Swimlanes for NCR / 8D drafting, failover, approval and write-back

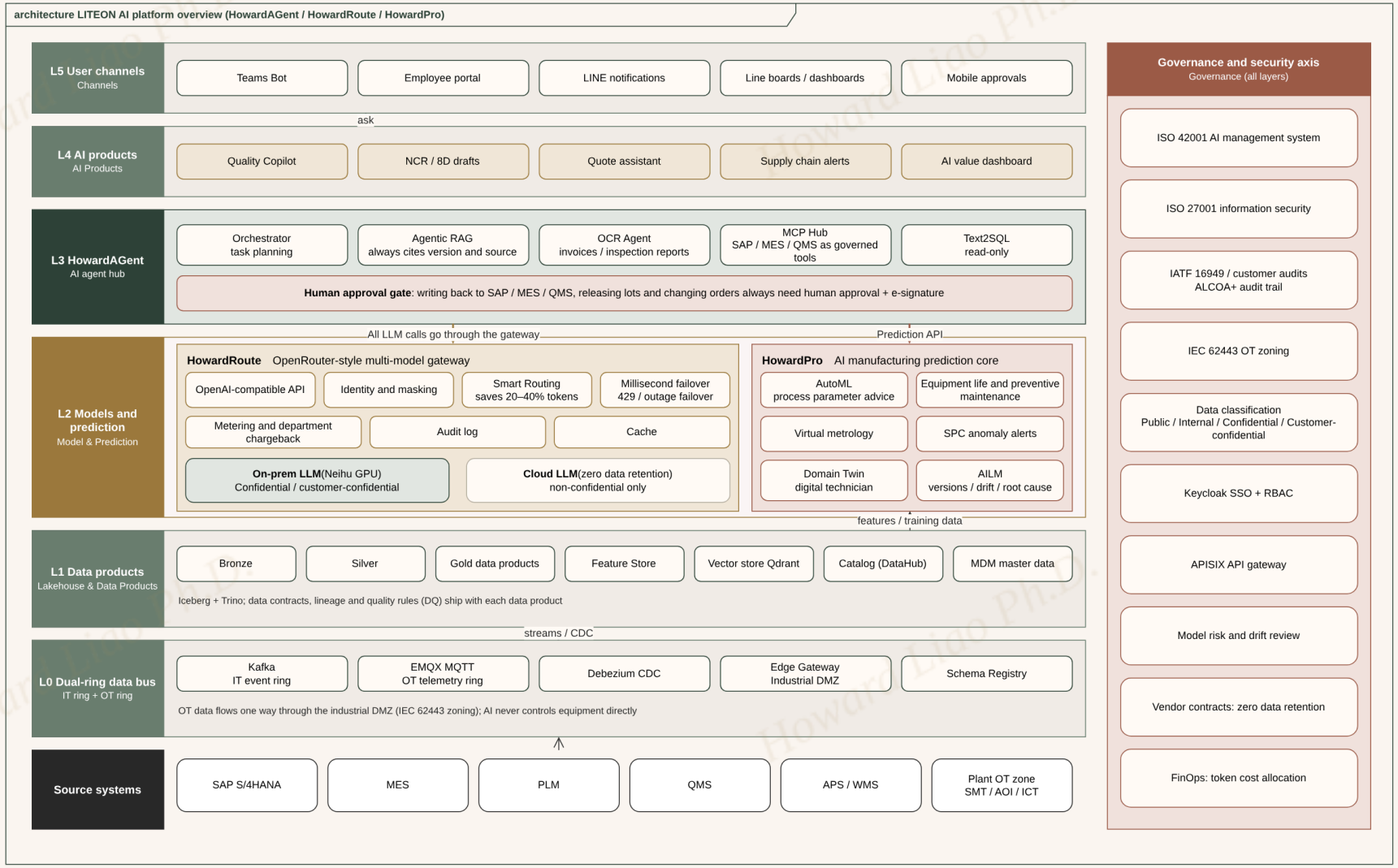


11 Sequence

NCR draft from question, routing and generation to approval and write-back

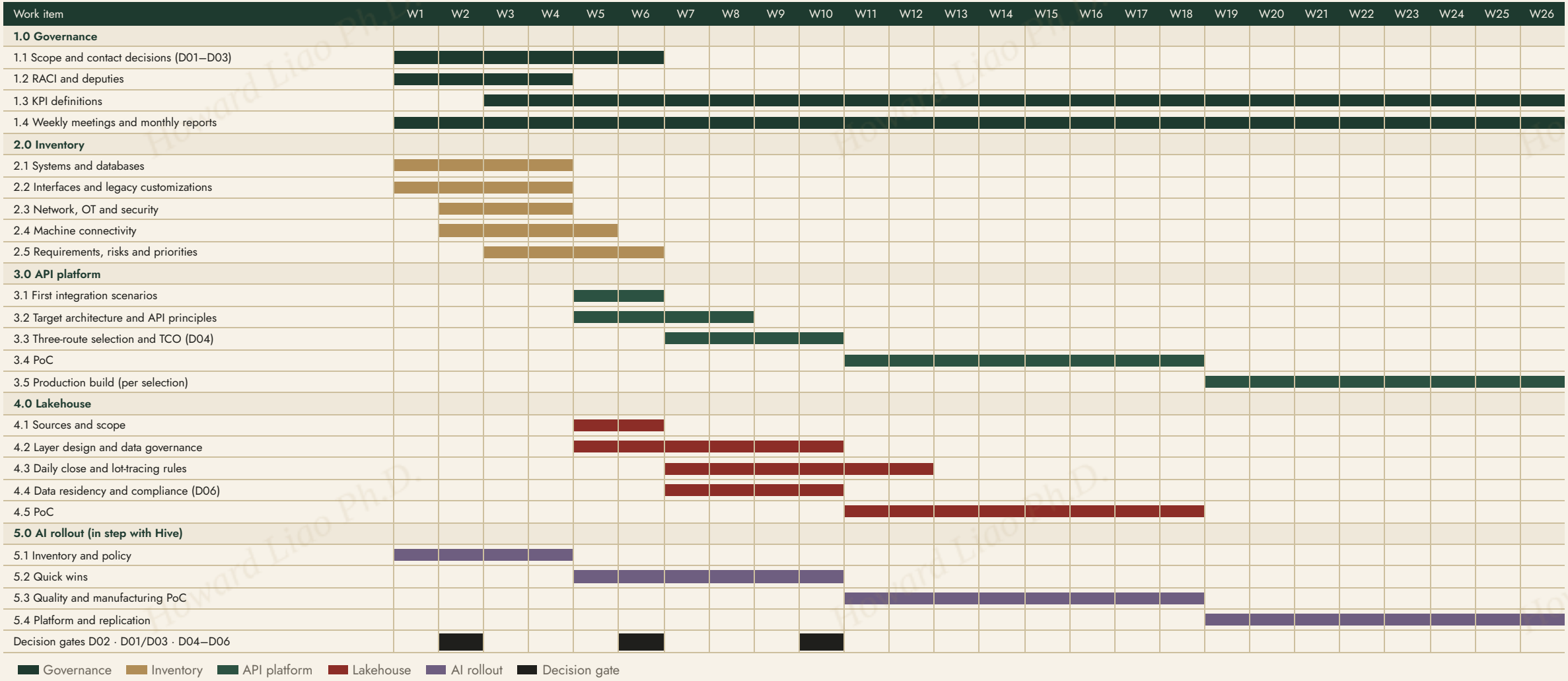
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LITEON AI platform: six layers plus a governance axis



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Project Hive: 26-week WBS Gantt chart



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Project Hive: RACI matrix

Task	CIO (sponsor)	AI platform lead	Infrastructure / OT	Applications (SAP / MES)	QA / compliance	BU PO / manufacturing	SI / vendor
1 Current-state and requirements inventory	A	R	C	C	C	C	—
2 API platform planning and build	I	A	C	R	I	C	R
3 Lakehouse planning and build	I	A	C	R	C	I	R
4 Platform selection and cost	A	R	C	C	C	I	C
5 Machine and system integration	I	C	R	A	I	R	R
6 Daily close and lot tracing	I	C	I	A	R	C	R
7 Network and security	I	C	A	C	C	C	R
8 PoC and stage acceptance	A	R	C	R	C	C	C
9 Delivery and operations roles	A	R	C	C	I	C	I

R responsible · A accountable · C consulted · I informed. Each task has exactly one A.

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Risk register: see the risks before talking speed

#	Risk	Impact	Mitigation	Owner
01	S/4HANA cut-over period	Integration built on a core in transition must be redone	Only inventory and knowledge use cases during the freeze; all interfaces go through the API gateway to reduce dependence on the core version	CIO / AI
02	Mixed machine brands and ages across plants	Connectivity cost and timeline grow	Inventory controller models; start with one line in one pilot plant	CIO / AI
03	Customer audits and IATF 16949	Records that fall short could affect customer certification	QA joins the PoC; acceptance includes audit trails and e-signatures	CIO / AI
04	Incomplete legacy inventory	Reports or interfaces missed after consolidation	Run old and new in parallel; retire only after complete data transfer	CIO / AI
05	Overloaded part-time staff	Daily support and the project compete	D02 fixes contacts and weekly hours; daily support comes first	CIO / AI
06	No owner for operations	Nobody responsible after go-live	D05 settled before the PoC, with the owner joining the PoC	CIO / AI
07	Competing for resources with the S/4HANA go-live	—	Before 2027 Q1, only use cases that do not touch SAP (quality, R and D knowledge)	AI
08	Insufficient data quality	—	Build data products and unified definitions first, then models	AI
09	PoC stuck at PoC	—	Define go-live criteria, PO and P and L line before starting	AI
10	Shadow AI	—	Provide a good official platform plus usage rules, rather than only prohibiting	AI
11	Runaway compute cost	—	Token/GPU cost allocated by BU, with quotas and monthly reports	AI

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Board KPI scorecard

Area	Metric	Target
Economic value	Verified annualized benefit (confirmed by the CFO)	Targets agreed within 60 days of joining
Platform	Number of live use cases; time to launch a new use case	≥ 10; ≤ 4 weeks
Adoption	Monthly active users of the AI platform as a share of employees	Rising every quarter
Organization	Certified POs; successors for technical leads	30; one per lead
Governance	Use-case risk assessment completion rate; major security incidents	100%; zero
Cost	Cost per thousand requests; compute budget attainment	Down every quarter; ±10%

Economic benefit targets are set with the CFO within 60 days of joining.

+15%
Product yield
Hongyan Technology | AI defect prediction

+18%
OEE
Hongyan Technology | Edge AIoT

−25%
IT operating cost
Hongyan Technology | System consolidation

−30%
Multi-cloud TCO
GameSparcs (6542) | FinOps

Zero downtime
Core services
GameSparcs (6542) | Platform migration

−40%
Delivery Lead Time
Global Internet Technology Group | AI code review

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Six decisions for the CIO

D01

Phase-one scope

Suggested: foundation only (API + lakehouse + first integrations); BI improvements in phase two. Blocks tasks 2, 3, 5, 6.

D02

Cross-functional contacts

Suggested: one dedicated contact each from infrastructure, applications, QA, manufacturing, finance and the pilot business group. Blocks task 1.

D03

Pilot plant and line

Suggested: one main line at the new Vietnam plant or Kaohsiung phase 2, with no legacy and representative data. Blocks tasks 5, 8.

D04

Budget and route

Suggested: approve the PoC budget first and the production budget after the PoC. Blocks task 8 and stage 5.

D05

Build and operations owner

Options: platform team, SI or a mix. Suggested: decide before the PoC and involve the owner. Blocks tasks 8, 9.

D06

Primary cloud region and data residency

Suggested: check customer contracts, regulations and national residency rules first, then choose Taiwan or a regional cloud. Blocks tasks 3, 4, 7.

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Media and keynote gallery



CIO IT Manager feature interview



CIO IT Manager, May 2024 issue



Keynote at a database vendor's Taipei conference



iThome feature



"Enterprise to the Cloud" industry event



e-Learning for civil servants: cloud management



ACE industry interview



iThome feature

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Certificates and completion records



Certificate of Profession

GS1 條碼管理師證書

This is to certify that

茲證明

LIAO LUN-HAO

廖倫豪

has successfully completed the requirements to be recognized as a GS1 Taiwan

通過 GS1 國際條碼認證考試，授與國際條碼管理師資格。

Certified Barcode Management Specialist

此證

CEO

財團法人中華民國商品條碼策進會

執行長 林 燁

Developed by GS1 Taiwan

Date Of Issue : 4/16/2015

Registration Number : 15041601002

發證日期：4/16/2015

證書字號：15041601002



ORACLE UNIVERSITY

CERTIFICATE OF COMPLETION

廖倫豪

HAS SUCCESSFULLY COMPLETED

Oracle EBS R12 TTT –Supply Chain Modules

(Essentials、OM、 PUR、 INV)

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ENROLLMENT ID

JOHN HALL SENIOR VICE PRESIDENT ORACLE CORPORATION

艾黛爾

INSTRUCTOR NAME

25-FEB-2012

DATE



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艾黛爾、何瑞強

INSTRUCTOR NAME

19-APR-2012

DATE



WBSA

WBSA CERTIFICATE

WORLD BUSINESS STRATEGIST ASSOCIATION

No.TWB 000030

Let it be known that on this 2007/08/31,

廖倫豪 Liao Lun Hao *has successfully become*

WBSA Marketing strategic planner.

WORLD BUSINESS STRATEGIST ASSOCIATION

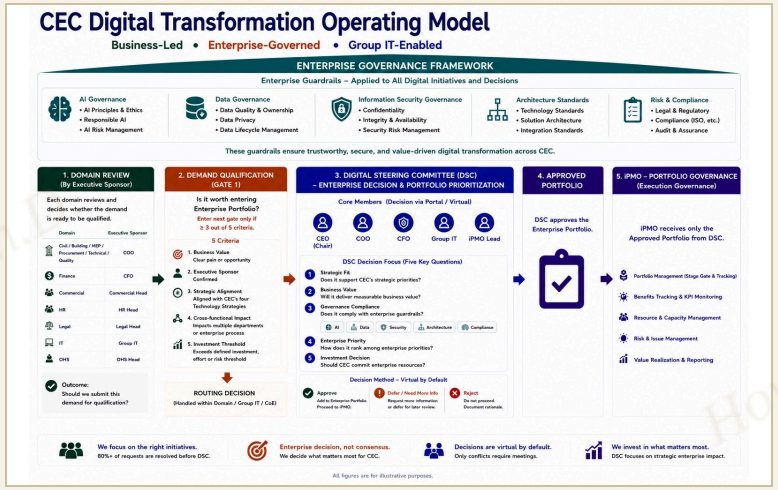
Chairman



台灣金融研訓院 EN 關於本院 課程 證照 就業招考 傳播/出版 研究 FE62 客戶服務 網站導覽 廖倫豪

台灣金融研訓院(TABF)的受訓相關記錄：

班級名稱	上課起日	上課訖日	課程時數	成績	出勤情況	考核結果	證書字號	操作
資訊安全意識、必備知識與實任E-Course(第2期)	2025/01/10	2025/04/10	2.00	90.00	全勤	合格	(114)金發證字第IS100100002962號 證書列印 雜字證書列印	
資安事件說明及預防措施E-Course(第2期)	2025/01/10	2025/04/10	2.50	90.00	全勤	合格	(114)金發證字第IS100110002962號 證書列印 雜字證書列印	
上市上櫃公司資通安全管控指引說明E-Course(第2期)	2025/01/10	2025/04/10	1.50	90.00	全勤	合格	(114)金發證字第IS100120002962號 證書列印 雜字證書列印	



CEC Digital Transformation Operating Model

Business-Led • Enterprise-Governed • Group IT-Enabled

ENTERPRISE GOVERNANCE FRAMEWORK

Enterprise Guardrails – Applied to All Digital Initiatives and Decisions

AI Governance
• AI Principles & Ethics
• Responsible AI
• AI Risk Management

Data Governance
• Data Quality & Ownership
• Data Privacy
• Data Lifecycle Management

Information Security Governance
• Confidentiality
• Integrity & Availability
• Security Risk Management

Architecture Standards
• Technology Standards
• Solution Architecture
• Integration Standards

Risk & Compliance
• Legal & Regulatory
• Compliance (ISO, etc.)
• Audit & Assurance

These guardrails ensure trustworthy, secure, and value-driven digital transformation across CEC.

1. DOMAIN REVIEW (By Executive Sponsor)

Each domain reviews and decides whether the demand is ready to be qualified.

Domain — Executive Sponsor
Civil / Building / MEP / Procurement / Technical / Quality — COO
Finance — CFO
Commercial — Commercial Head
HR — HR Head
Legal — Legal Head
IT — Group IT
QHSE — QHSE Head

Outcome: Should we submit this demand for qualification?

2. DEMAND QUALIFICATION (GATE 1)

Is it worth entering Enterprise Portfolio?
Enter next gate only if ≥ 3 out of 5 criteria.

5 Criteria
1. Business Value: Clear pain or opportunity
2. Executive Sponsor: Confirmed
3. Strategic Alignment: Aligned with CEC's four Technology Strategies
4. Cross-functional Impact: Impacts multiple departments or enterprise process
5. Investment Threshold: Exceeds defined investment, effort or risk threshold

ROUTING DECISION
(Handled within Domain / Group IT / CoE)

3. DIGITAL STEERING COMMITTEE (DSC) – ENTERPRISE DECISION & PORTFOLIO PRIORITIZATION

Core Members (Decision via Portal / Virtual)
CEO (Chair) · COO · CFO · Group IT · IPMO Lead

DSC Decision Focus (Five Key Questions)
1. Strategic Fit: Does it support CEC's strategic priorities?
2. Business Value: Will it deliver measurable business value?
3. Governance Compliance: Does it comply with enterprise guardrails?
4. Enterprise Priority: How does it rank among enterprise priorities?
5. Investment Decision: Should CEC commit enterprise resources?

Decision Method – Virtual by Default
Approve · Defer / Need More Info · Reject

4. APPROVED PORTFOLIO

DSC approves the Enterprise Portfolio.

5. IPMO – PORTFOLIO GOVERNANCE (Execution Governance)

IPMO receives only the Approved Portfolio from DSC.

Portfolio Management (Stage Gate & Tracking)
Benefits Tracking & KPI Monitoring
Resource & Capacity Management
Risk & Issue Management
Value Realization & Reporting

We focus on the right initiatives. 80%+ of requests are resolved before DSC.
Enterprise decision, not consensus. We decide what matters most for CEC.
Decisions are virtual by default. Only conflicts require meetings.
We invest in what matters most. DSC focuses on strategic enterprise impact.

All figures are for illustrative purposes.



Official portrait

Howard Liao, *Ph.D.*

廖 倫 豪 博 士

CIO · CISO · ENTERPRISE AI ARCHITECT

Enterprise AI platform and digital transformation leader | VP of Technology / CIO / CISO

Howard Liao Ph.D.



Grazie

Thank you | I look forward to meeting in person

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Howard Liao Ph.D.